PRERANA KULLA

Charlotte, NC
kprerana@unc.edu
(704) 806-3408
www.linkedin.com/in/prerana-kulla
https://preranakulla.github.io/Portfolio-Website/

EDUCATION

University of North Carolina at Chapel Hill – Chapel Hill, NC Fall 2025 – Spring 2029

- *Psychology, Computer Science, Business of Health (Minor)*

North Carolina School of Science and Mathematics – Morganton, NC May 2025

High School Diploma (4.0 UW, 5.0 W GPA)

RESEARCH EXPERIENCE

Examining the effects of estradiol on neural and molecular response to reward in perimenopausal onset anhedonia and psychosis | UNC School of Medicine, Center for Women's Mood Disorders | clinical recruitment and screening, Excel, data presentation Fall 2025 – Current

Mentors: Reese Gray, B.S., Dr. Crystal Schiller

- Collaborate with members of the lab on data collection via brain scans and blood draws. Screen for control and main group participants for study recruitment. Practice data collection and file organization.

Utilizing Enzymatic and Chemical Delignification Methods on Fibrous Plants for Decentralized Production of Absorbent Materials for Menstrual Pads in Semi-Arid Regions | Decortication, enzymatic delignification using laccase enzymes, autoclaving, FTIR spectroscopy, cryogenic grinding, microscopy using Leica ICC50 W, absorbency testing April 2024 – March 2025

Mentor: Mrs. Jennifer Williams

- Collected and analyzed datasets on cellulose, hemicellulose, lignin, and moisture content to build a multivariate polynomial regression model that predicted the most suitable fibrous leaf substitute for cotton. Based on the results, utilized delignification and mechanical fluffing methods to create plants from pineapple and sisal fiber from leaves.

Kulla, P. (2024). Utilizing Enzymatic and Chemical Delignification Methods on Fibrous Plants for Decentralized Production of Absorbent Materials for Menstrual Pads in Semi-Arid Regions. The Morganton Scientific.

Design of Supramolecular Nanofibers for Gonorrhea Vaccines | Duke Research in Engineering Program | Liberty Blue, HPLC, MALDI-TOF, TEM, Zeta Potential June 2023 – August 2023

Mentors: Dr. Joel H. Collier, Emily F. Roe

- Synthesized supramolecular nanofibers for sublingual gonorrhea vaccines, involving peptide synthesis purification, nanofiber characterization, and mucoadhesion analysis, contributing to the development of heat-stable, mucosal vaccine platforms.

LEADERSHIP AND COMMUNITY INVOLVMENT

Strive2Thrive Edu | Charlotte, NC | Tutor & Project Coordinator March 2021 – Present

- Instructed students in Uganda and Zambia on computing basics and biology, adapting curriculum for diverse learning environments.
- Facilitated seamless communication between tutors and students to optimize educational delivery and administrative efficiency.

MediLink | Charlotte, NC | Founder April 2023 – Present

- Established a non-profit organization bridging the gap between business and healthcare to equip aspiring entrepreneurs with industry-specific leadership skills.
- Collaborate with cross-functional partners to develop training modules focused on medical-financial literacy and social impact.

Menstrual Movement in Morganton | Morganton, NC | Founder May 2024 – May 2025

- Lead donation drives that secured menstrual hygiene products for 4,000+ women across Harrisburg (US), Hyderabad (India), and Morganton (US).

- Published Red Ribbon, an educational resource for adolescent health, and conducted outreach to 200+ students via school visits and webinars.

WORK EXPERIENCE

UNC ResNET IT Services – Chapel Hill, NC
Residential Computing Consultant

- Provide in-room technical support for UNC residential students by troubleshooting r
  setup, and CCI laptop issues.

Kumon – Harrisburg, NC
Tutor

- Provided comprehensive tutoring in mathematics and reading to students across vari
  their academic growth and fostering a positive learning environment.

ADDITIONAL INFORMATION

Certifications: CPR, EMT-B
Relevant Coursework: Analytical Chemistry, Calculus of Functions of One Variable II, Cellular Biology, Sociological Perspectives, Introduction to Programming, General Psychology, Introduction to Data Models and Inference, Introduction to Business: People, Profits, Planet
Computer Skills: Python, Java, Microsoft Office, Google Workspace
Languages: English (native fluency), Telugu (native fluency), Hindi (conversational fluency), Spanish (intermediate)
Campus Involvement: CPALS, Youth Movement Against Alzheimer's, Carolina Women in Business, Technology Consulting Club, NEX GEN USA, Honors Carolina Ambassador
Honors/Awards: Accelerated Research Program, Honors Carolina, Morehead-Cain Finalist, North Carolina Student Academy of Science 2nd Place (2024, 2025), University of South Carolina Top Scholar Recipient